GG-B-89b
MARCH 4, 1958
SUPERSEDING
Int. Fed. Spec. GG–B–0089a (GSA–FSS)
June 18, 1957 and
Fed. Spec. GG–B–89
August 5, 1948

FEDERAL SPECIFICATION

BAND, ABRASIVE (SILICON-CARBIDE), FOR DENTAL LATHES

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers silicon-carbide bands suitable for rough finishing and trimming of acrylic resins. Bands shall be for use with arbor chucks which fit dental laboratory lathes.

1.2 Classification.

1.2.1 *Classes and sizes.*—Abrasive bands shall be of the following classes and sizes, as specified (see 6.2).

Class 1—60 grit, for coarse work.
Class 2—80 grit, for medium work.
Class 3—120 grit, for fine work.
Size 1—0.25-inch inside diameter.
Size 2—0.375-inch inside diameter.
Size 3—0.50-inch inside diameter.
Size 4—0.762-inch inside diameter.

1.2.2 *Assortment.* — Assortments of the various classes and sizes shall be as specified (see 6.2).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

RR–S–366—Sieves, Standards for Testing Purposes.
CCC–J–191—Jean; Bleached.
PPP–B–585—Boxes; Wood, Wirebound.
PPP–B–601—Boxes, Wood, Cleated-Plywood.
PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
PPP–B–676 — Boxes, Set-Up, Paperboard.

Federal Standard:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, At-

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lanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL–B–10377—Box, Wood, Cleated, Veneer, Paper Overlaid.

MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Silicon carbide.*—The silicon carbide shall be as hereinafter specified.

3.1.2 *Jeans cloth.*—The jeans cloth shall conform to the requirements of Federal Specification CCC–J–191.

3.2 Construction.—The finished bands shall consist of an inner and outer band fastened together in the shape of a closed circular band. The two concentric bands shall be fastened together by means of a suitable adhesive.

3.2.1 *Outer band.*—The outer band shall be jeans cloth and shall be covered on its outer surface with silicon carbide.

3.2.2 *Inner band.*—The inner band shall be jeans cloth.

3.3 Width.—The bands shall be not less than 0.50 inch wide.

3.4 Diameter.—The diameter shall be as shown in 1.2.1, except that a minus tolerance of 0.005 inch will be permissible.

3.5 Joints.—The joint of the inner band shall be at a point opposite the joint of the outer abrasive band, and shall be butt type at an angle of 45°.

3.6 Fineness.—The size of the abrasive grain for the various classes shall be as hereinafter specified. The sieves used shall be U. S. standard sieves conforming to the requirements of Federal Specification RR–S–366.

3.6.1 *Class 1.*—The size of the abrasive grain shall be No. 60 standard mesh. When tested as described in 4.3.1, not more than 3 percent of the abrasive shall pass a No. 80 sieve, and 100 percent shall pass a No. 40 sieve. Not less than 45 percent of the abrasive which passes a No. 50 sieve shall be retained on a No. 60 sieve, and a total of not less than 65 percent shall be retained on No. 60 and No. 70 sieves.

3.6.2 *Class 2.*—The size of the abrasive grain shall be No. 80 standard mesh. When tested as described in 4.3.1, not more than 3 percent of the abrasive shall pass a No. 120 sieve, and 100 percent shall pass a No. 60 sieve. Not less than 40 percent of the abrasive which passes a No. 70 sieve shall be retained on a No. 80 sieve, and a total of not less than 70 percent shall be retained on No. 80 and No. 100 sieves.

3.6.3 *Class 3.*—The size of the abrasive grain shall be No. 120 standard mesh. When tested as described in 4.3.1, not more than 3 percent of the abrasive shall pass a No. 230 sieve, and 100 percent shall pass a No. 100 sieve. Not less than 30 percent of the abrasive which passes a No. 120 sieve shall be retained on a No. 140 sieve, and a total of not less than 60 percent shall be retained on No. 140 and No. 170 sieves.

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3.7 Adhesive.—The adhesive shall be of good quality and suitable for the purpose intended. When tested as described in 4.3.2, the inner and outer bands shall not separate.

3.8 Flexibility.—The bands will be of such flexibility as to withstand the required operating conditions. When tested as described in 4.3.3, there shall be no cracking, curling, or loss of abrasive.

3.9 Workmanship.—The finished band shall be first class in every respect. Band shall be uniformly covered with abrasive and shall be free from any imperfections which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Sampling.—Sampling procedures shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105. In addition, samples may be requested by the procuring activity for testing at Government laboratory.

4.2 Inspection.—Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance. Final inspection of the finished article shall be made either at point of production or at point of delivery, as specified (see 6.2). In case of factory inspection, every facility shall be afforded inspectors, by the manufacturer, for the prosecution of their work.

4.3 Tests.

4.3.1 *Fineness.*—A 100-gram sample shall be shaken for 5 minutes on a Ro-tap machine, or equivalent, vibrating at a speed of 290 revolutions per minute and 156 taps per minute. Where more than a single sieve is used, they shall be agitated simultaneously.

4.3.2 *Adhesiveness.*—Two bands shall be submerged in distilled water at 70° ± 2° Fahrenheit for 5 ± ½ minutes.

4.3.3 *Flexibility.*—Two bands shall be cut open at the joint of the outer band. The specimens shall be flexed over a smooth rod (abrasive away from rod) 9/16 inch in diameter and shall be pulled (with the ends parallel) over the rod 10 times. The specimens shall be inspected for evidence of cracking, curling, or flaking of the coating from the cloth backing as a result of the test.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Unit of issue.*—One hundred bands, as specified, constitute one unit.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table I shall be packaged in each unit and exterior container, as applicable, for the required level of protection specified in procurement document.

TABLE I.—*Packaging quantities*

Federal stock number	Unit package	Intermediate package	Shipping container
6520–501–5150	1 unit	Not required	96 units

5.1.2.1 *Packing variation permitted.*—If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

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5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Each unit shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–676. Each carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.*—Unless otherwise specified, bands shall be packaged in accordance with accepted commercial practice.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be designed for type 2 load and constructed in accordance with Federal Specifications PPP–B–585, style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, table III; or Military Specification MIL–B–10377, table II.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall conform to Federal Specification PPP–B–601, table I, using type III, class 1 plywood, or shall be designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 2 use; PPP–B–621, table I, of Military Specification MIL–B–10377, table I.

5.2.3 *Level C.*—The bands shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking.

5.3.1 *Unit package.*—Each unit package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2 *Shipping container.*—Shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Class and size required (see 1.2.1).
(*c*) Assortment required (1.2.2).
(*d*) Inspection required (see 4.2).
(*e*) Place of final inspection (see 4.2).
(*f*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Intended use.—The bands covered by this specification are intended for use on dental lathe abrasive band chucks for the purpose of cutting and finishing dentures and for general laboratory use.

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